Module End Assignment:

“Data Cleaning, Analysis & Visualization with Excel”

Copied the given raw data Customer_Dim, Product_Dim, Store_Dim and Sales_Fact into a new sheets

Used **Autofit Row Height and Autofit Column Width** options

Converted the data into table and used **Borders**

CUSTOMER_DIM :

STEP 1:

Found the **missing values in Loyalty Level column** in Customer_Dim sheet and used Mode to fill the missing values

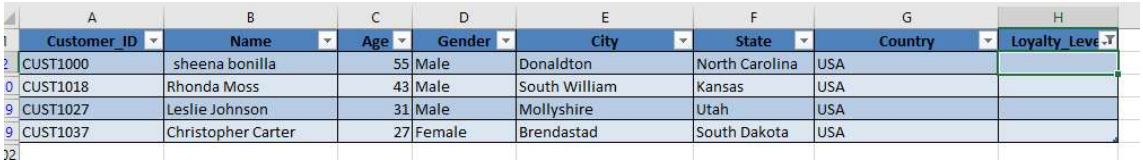
Used **=COUNTIF** to calculate the number of Bronze,Gold,Platinum and Gold in Loyalty Level column and found Platinum appears most compared to others ,so decided to fill the blank values with **Platinum**



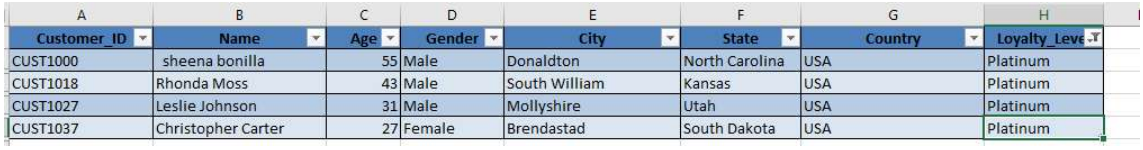
M7 =COUNTIF(Table1[Loyalty_Level],"Platinum")

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Leve
CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA	
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver

No.of Bronze Loyalty Level	125
No.of Platinum Loyalty Level	145
No.of Gold Loyalty Level	122
No.of Silver Loyalty Level	104



Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Leve
CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA	
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	
CUST1037	Christopher Carter	27	Female	Brendastad	South Dakota	USA	



Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Leve
CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Platinum
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	Platinum
CUST1037	Christopher Carter	27	Female	Brendastad	South Dakota	USA	Platinum

STEP 2:

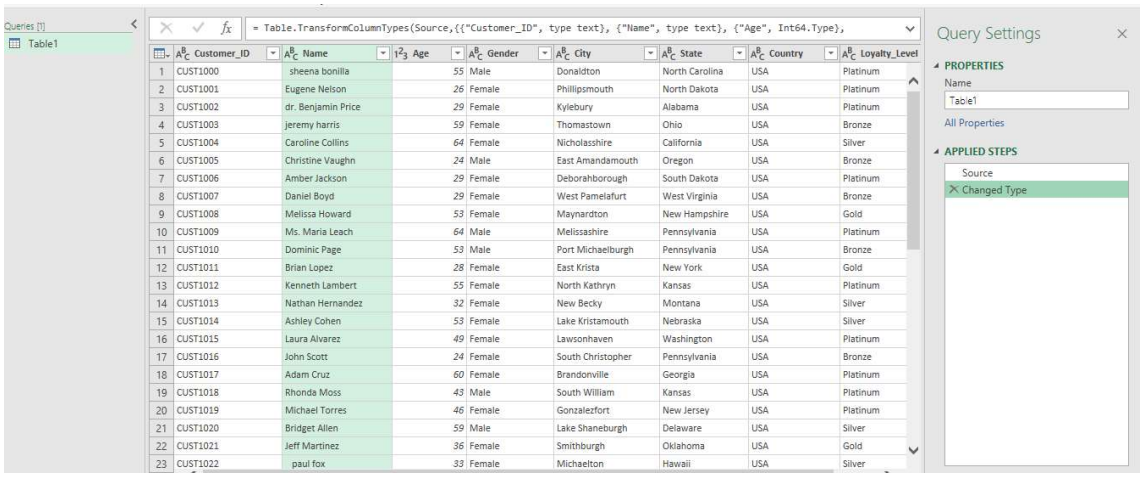
Found the inconsistent data in the Customer ID column and corrected it by comparing it with the given data



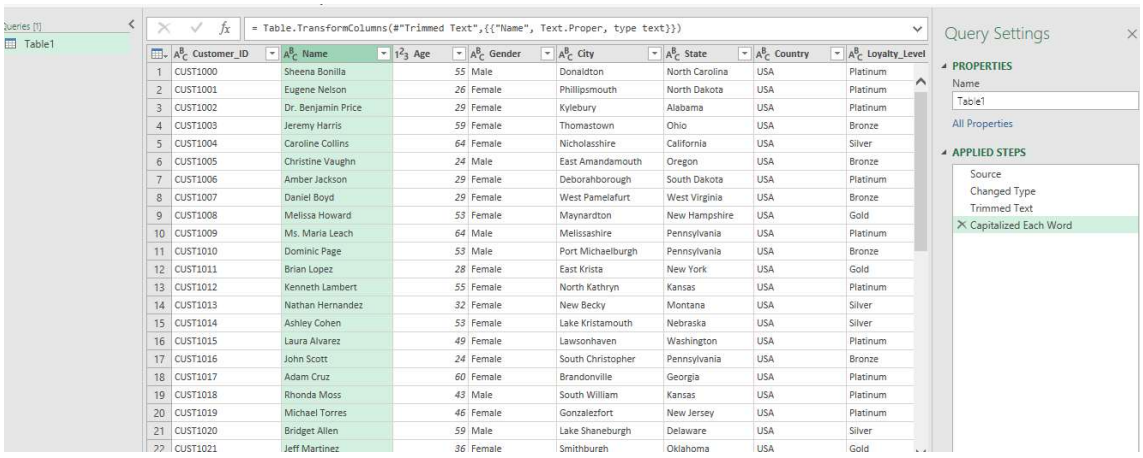
Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Leve
C-1464	David Pierce	45	Female	Grayland	Washington	USA	Platinum
C-1486	Eric Lee	22	Female	Lake April	Vermont	USA	Bronze
C-1491	Kaitlin Brown	52	Female	Kellyberg	South Dakota	USA	Gold
C-1499	Philip Pennington	27	Female	East Valerieview	Oregon	USA	Platinum

	A	B	C	D	E	F	G	H
1	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Leve
466	CUST1464	David Pierce	45	Female	Grayland	Washington	USA	Platinum
488	CUST1486	Eric Lee	22	Female	Lake April	Vermont	USA	Bronze
493	CUST1491	Kaitlin Brown	52	Female	Kellyberg	South Dakota	USA	Gold
501	CUST1499	Philip Pennington	27	Female	East Valerieview	Oregon	USA	Platinum
502								

STEP 3:
Imported the excel data into a Power query and used **trim & capitalize each word** option on **Name** column

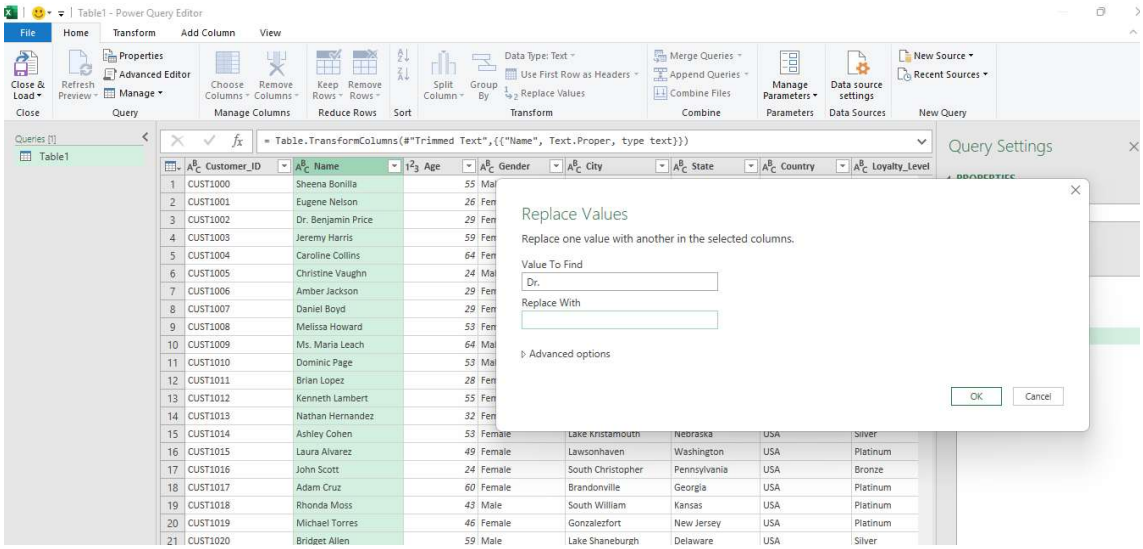


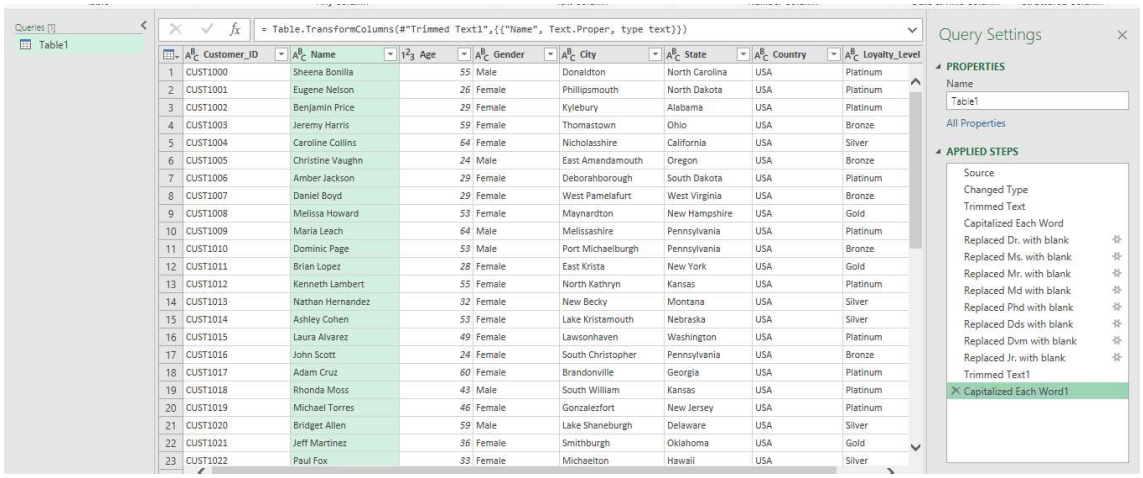
	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
1	CUST1000	sheena bonilla	55	Male	Donaldton	North Carolina	USA	Platinum
2	CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
3	CUST1002	dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
4	CUST1003	jeremy harris	59	Female	Thomastown	Ohio	USA	Bronze
5	CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
6	CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
7	CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
8	CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
9	CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
10	CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
11	CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
12	CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
13	CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
14	CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
15	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
16	CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
17	CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
18	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
19	CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Platinum
20	CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
21	CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
22	CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
23	CUST1022	paul fox	33	Female	Michaelton	Hawaii	USA	Silver



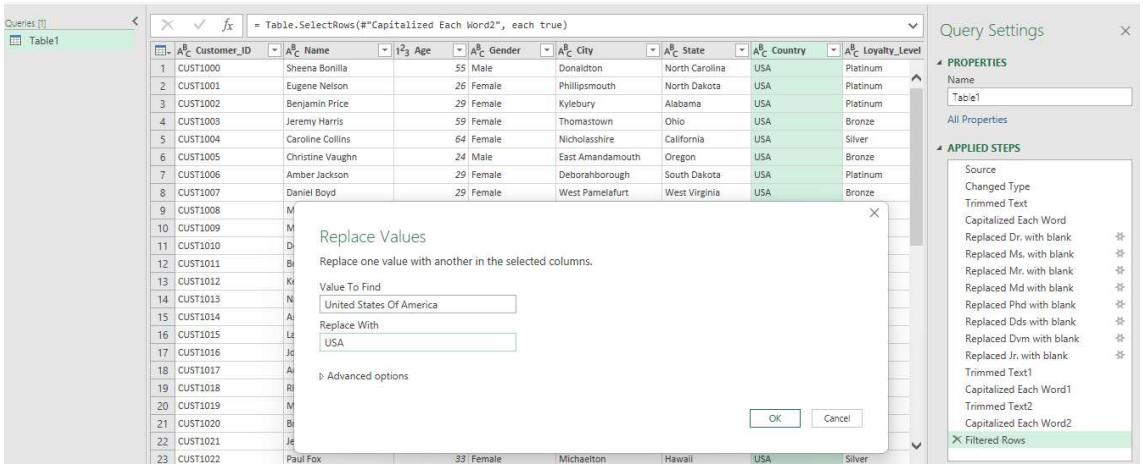
	Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
1	CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA	Platinum
2	CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
3	CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
4	CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
5	CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
6	CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
7	CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
8	CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
9	CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
10	CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
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14	CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
15	CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
16	CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
17	CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
18	CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
19	CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Platinum
20	CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
21	CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
22	CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold

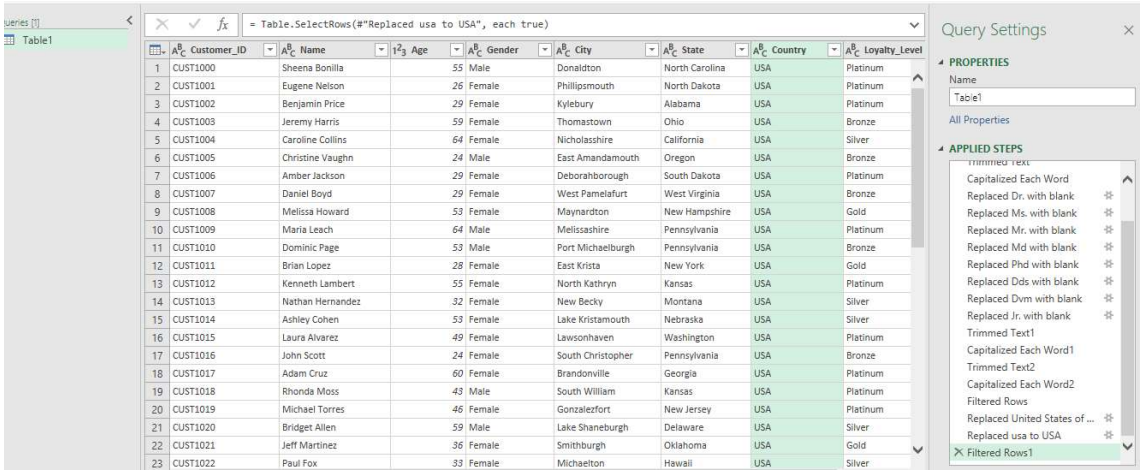
Noticed **Dr. , Mr. & Ms.** prefixes on few names and **md,Jr. , Dv, Phd and Dds** suffices ,removed them by replacing with blank values ,then trimmed again and capitalized each word





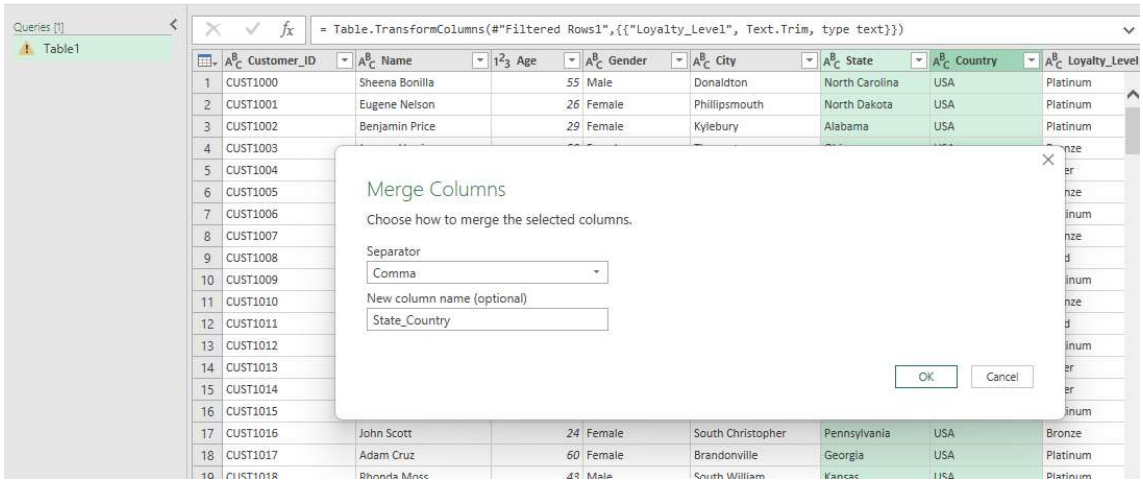
Trimmed and capitalized each word in City and State column
Replaced **United States Of America and usa to USA i**n Country column

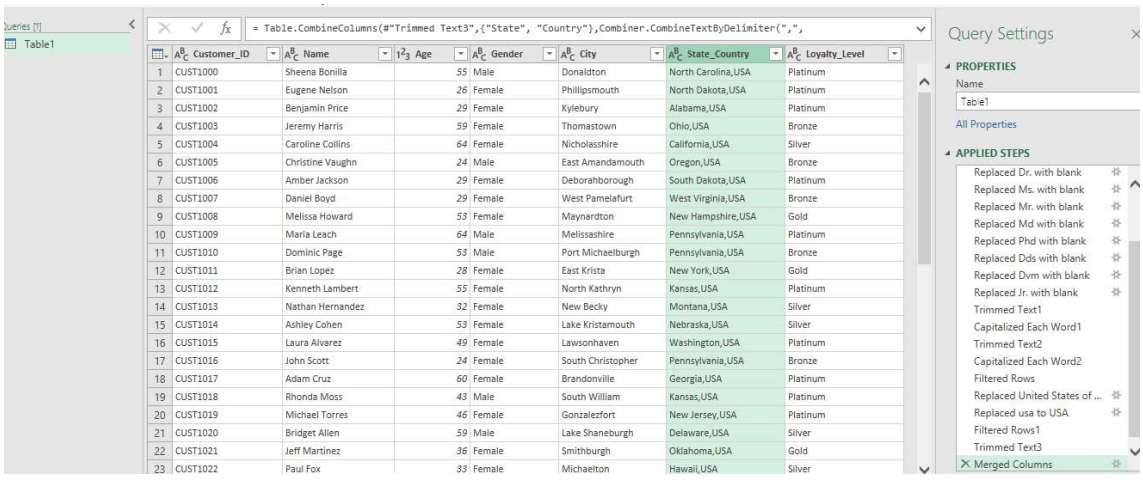




STEP 4:

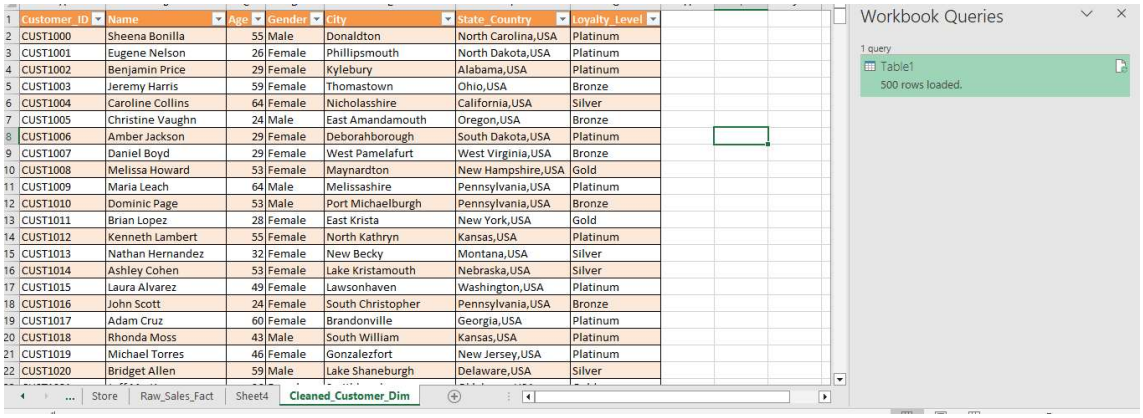
Merged a state and country column to a single column named **State_country**





STEP 5:

Loaded the cleaned data to a excel



Customer_ID	Name	Age	Gender	City	State_Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina,USA	Platinum
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota,USA	Platinum
CUST1002	Benjamin Price	29	Female	Kylebury	Alabama,USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio,USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California,USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon,USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota,USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia,USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire,USA	Gold
CUST1009	Maria Leach	64	Male	Melissashire	Pennsylvania,USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania,USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York,USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas,USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana,USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska,USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington,USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania,USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia,USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas,USA	Platinum
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey,USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware,USA	Silver

Workbook Queries
1 query
Table1
500 rows loaded.

Store | Raw_Sales_Fact | Sheet4 | Cleaned_Customer_Dim

PRODUCT_DIM:

STEP 1:

Found the inconsistent data in Product ID column and corrected it



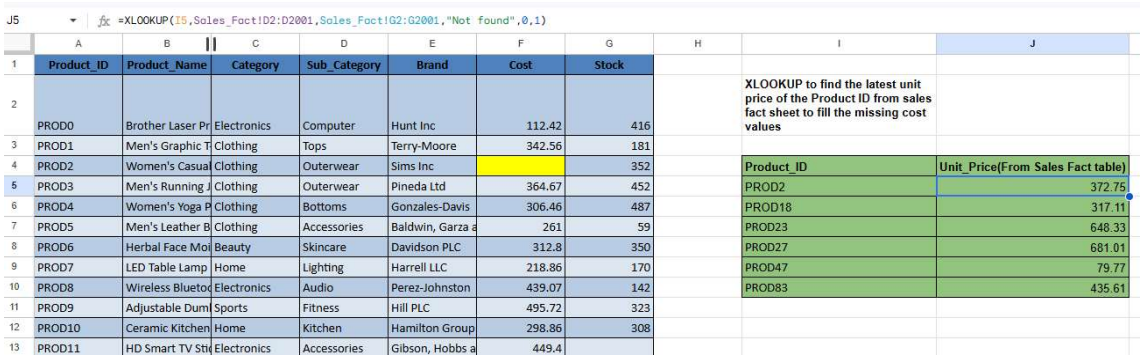
Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
P-3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
P-16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256



Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD3	Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD16	Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	256

STEP 2:

Filled the missing values in the Cost column by using **XLOOKUP** ,Sales_Fact sheet has multiple unit prices for each product id, so sorted based on date and used Xlookup to fill the latest unit price in the missing cost column



=XLOOKUP(I5,Sales_Fact!D2:D2001,Sales_Fact!G2:G2001,"Not found",0,1)

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD0	Brother Laser Pr	Electronics	Computer	Hunt Inc	112.42	416
PROD1	Men's Graphic T	Clothing	Tops	Terry-Moore	342.56	181
PROD2	Women's Casual	Clothing	Outerwear	Sims Inc		352
PROD3	Men's Running J	Clothing	Outerwear	Pineda Ltd	364.67	452
PROD4	Women's Yoga P	Clothing	Bottoms	Gonzales-Davis	306.46	487
PROD5	Men's Leather B	Clothing	Accessories	Baldwin, Garza a	261	59
PROD6	Herbal Face Moi	Beauty	Skincare	Davidson PLC	312.8	350
PROD7	LED Table Lamp	Home	Lighting	Harrell LLC	218.86	170
PROD8	Wireless Bluetoo	Electronics	Audio	Perez-Johnston	439.07	142
PROD9	Adjustable Dum	Sports	Fitness	Hill PLC	495.72	323
PROD10	Ceramic Kitchen	Home	Kitchen	Hamilton Group	298.86	308
PROD11	HD Smart TV Sti	Electronics	Accessories	Gibson, Hobbs a	449.4	

XLOOKUP to find the latest unit price of the Product ID from sales fact sheet to fill the missing cost values

Product_ID	Unit_Price(From Sales Fact table)
PROD2	372.75
PROD18	317.11
PROD23	648.33
PROD27	681.01
PROD47	79.77
PROD83	435.61

Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock
PROD2	Women's Casual Hoodie	Clothing	Outerwear	Sims Inc	372.75	352
PROD18	Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix	317.11	343
PROD23	Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos	648.33	360
PROD27	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	681.01	378
PROD47	Makeup Brush Set	Beauty	Makeup	Cook Inc	79.77	431
PROD83	Hydrating Face Cleanser	Beauty	Skincare	Butler Group	435.61	159

STEP 3:
Filtered based on category & sub category and used **Median** (as stocks may have huge variations used median instead of mean)to fill the missing stock values



J102 =MEDIAN(J15:J92)

	A	B	C	D	E	F	G	H	I	J
1	Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock			
13	PROD11	HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	375			
15	PROD13	USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	453			453
44	PROD42	Smart Home WiFi Camera	Electronics	Accessories	Holmes LLC	450.71	389			361
50	PROD48	Car Dashboard Phone Holder	Electronics	Accessories	Byrd, Stewart and Valdez	177.98	103			361
53	PROD51	HDMI Cable – 2m	Electronics	Accessories	Bautista, Green and Hale	88.49	246			103
69	PROD67	Smart LED Light Strip	Electronics	Accessories	Flores-Davis	349.17	476			453
92	PROD90	Gaming Mouse Pad XL	Electronics	Accessories	Sharp, Taylor and Rodriguez	340.14	361			361
102									MEDIAN	375

STORE_DIM:
Converted the data in store_dim to a table

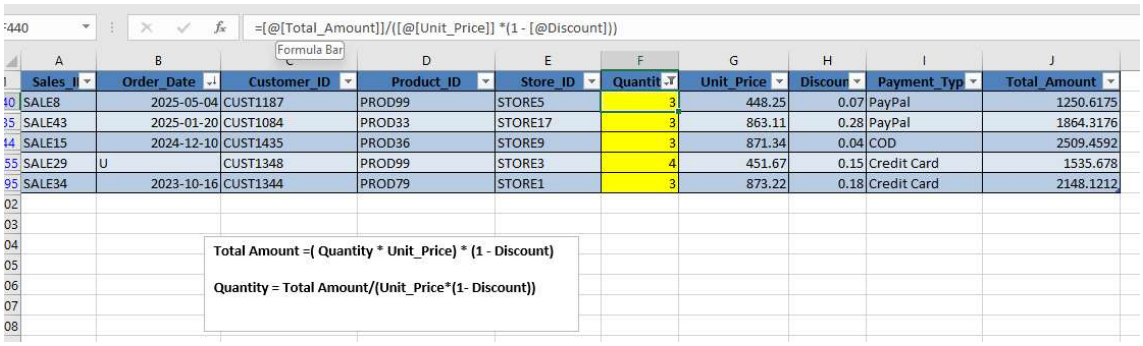
	A	B	C	D	E
1	Store_ID	Store_Name	Region	City	Store_Type
2	STORE0	Rivera-Warner	East	Washingtonfurt	Flagship
3	STORE1	Martinez, Adkins and Johnson	North	Danielmouth	Flagship
4	STORE2	Black-Horn	South	North Nancyview	Outlet
5	STORE3	Johnson-Wright	North	Hoodberg	Outlet
6	STORE4	Alexander Group	South	Port Sharon	Online
7	STORE5	Gray Ltd	North	New Michaelchester	Outlet
8	STORE6	Benson-Elliott	East	Port Josephberg	Online
9	STORE7	Martinez-Mullins	South	Williamsfurt	Online
10	STORE8	Bean, Jones and Park	West	Thompsonborough	Online
11	STORE9	Moore, Padilla and Butler	South	East Johnstad	Flagship
12	STORE10	Hill, Wagner and Ballard	East	East Davidville	Online
13	STORE11	Aguirre-Keller	South	Lake Michael	Online
14	STORE12	Thomas, Bridges and Harris	North	Sabrinatown	Online
15	STORE13	Gardner, Humphrey and Contreras	East	Johnsonville	Flagship
16	STORE14	Mcdonald Inc	South	Christopherland	Outlet
17	STORE15	Shelton PLC	North	East Jenniferstad	Flagship
18	STORE16	Frederick-Wright	West	New Jimmymouth	Flagship
19	STORE17	Holloway Ltd	North	Bensonhaven	Flagship
20	STORE18	Guerrero-Schmitt	East	East Stevechester	Flagship
21	STORE19	Mann-Austin	East	Vasquezborough	Online

SALES_FACT:
STEP 1:
Found few missing values in the Quantity column and used the below formula to calculate the missing value

Quantity = Total Amount / (Unit_Price*(1- Discount))



	A	B	C	D	E	F	G	H	I	J
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discour	Payment_Typ	Total_Amount
40	SALE8	2025-05-04	CUST1187	PROD99	STORE5		448.25	0.07	PayPal	1250.6175
35	SALE43	2025-01-20	CUST1084	PROD33	STORE17		863.11	0.28	PayPal	1864.3176
44	SALE15	2024-12-10	CUST1435	PROD36	STORE9		871.34	0.04	COD	2509.4592
55	SALE29	2024-01-11	CUST1348	PROD99	STORE3		451.67	0.15	Credit Card	1535.678
95	SALE34	2023-10-16	CUST1344	PROD79	STORE1		873.22	0.18	Credit Card	2148.1212
02										



:440 =[@[Total_Amount]]/([@[Unit_Price]] *(1 - [@Discount]))

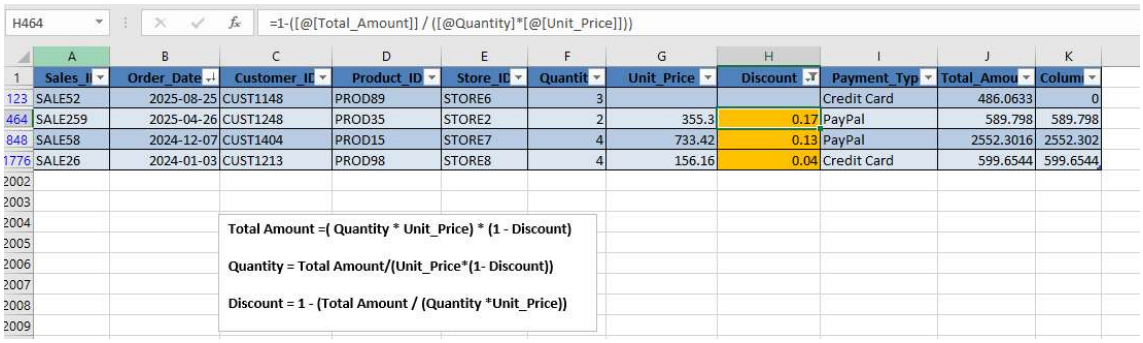
Formula Bar

	A	B	C	D	E	F	G	H	I	J
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discour	Payment_Typ	Total_Amount
40	SALE8	2025-05-04	CUST1187	PROD99	STORE5	3	448.25	0.07	PayPal	1250.6175
35	SALE43	2025-01-20	CUST1084	PROD33	STORE17	3	863.11	0.28	PayPal	1864.3176
44	SALE15	2024-12-10	CUST1435	PROD36	STORE9	3	871.34	0.04	COD	2509.4592
55	SALE29	U	CUST1348	PROD99	STORE3	4	451.67	0.15	Credit Card	1535.678
95	SALE34	2023-10-16	CUST1344	PROD79	STORE1	3	873.22	0.18	Credit Card	2148.1212
02										
03										
04										
05										
06										
07										
08										

Total Amount =(Quantity * Unit_Price) * (1 - Discount)

Quantity = Total Amount/(Unit_Price*(1- Discount))

STEP 2:
Filled the missing discount values using formula



H464 =1-([@[Total_Amount]] / ([@Quantity]*[@[Unit_Price]]))

	A	B	C	D	E	F	G	H	I	J	K
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discount	Payment_Typ	Total_Amou	Colum
123	SALE52	2025-08-25	CUST1148	PROD89	STORE6	3			Credit Card	486.0633	0
464	SALE259	2025-04-26	CUST1248	PROD35	STORE2	2	355.3	0.17	PayPal	589.798	589.798
848	SALE58	2024-12-07	CUST1404	PROD15	STORE7	4	733.42	0.13	PayPal	2552.3016	2552.302
1776	SALE26	2024-01-03	CUST1213	PROD98	STORE8	4	156.16	0.04	Credit Card	599.6544	599.6544
2002											
2003											
2004											
2005											
2006											
2007											
2008											
2009											

Total Amount =(Quantity * Unit_Price) * (1 - Discount)

Quantity = Total Amount/(Unit_Price*(1- Discount))

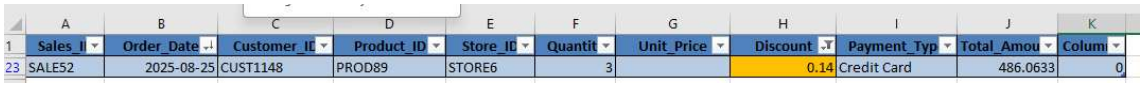
Discount = 1 - (Total Amount / (Quantity *Unit_Price))

1 row has both unit price and discount missing , so calculated descriptive statistics using **Analysis Toolpak addin** and filled the missing discount value with **Median**

Missing values in the Discount column were imputed using the median discount value. Median was chosen because it is less affected by extreme values and is suitable for numerical discount data ranging from 0 to 0.3



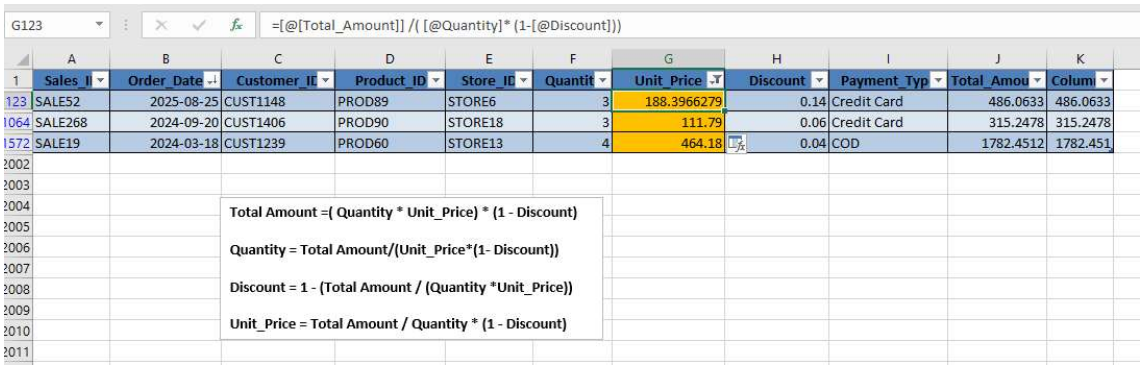
	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discount	Payment_Typ	Total_Amou	Colum		
2	SALE214	2025-10-13	CUST1199	PROD48	STORE16	1	775.84	0.26	Credit Card	574.1216	574.1216		
3	SALE504	2025-10-13	CUST1167	PROD47	STORE2	3	843.44	0.22	Credit Card	1973.6496	1973.65	Using Analysis Toolpak addin	
4	SALE592	2025-10-13	CUST1388	PROD92	STORE17	3	259.74	0.24	Credit Card	592.2072	592.2072	Mean	0.148086172
5	SALE1457	2025-10-13	CUST1238	PROD43	STORE18	4	975.47	0.04	Credit Card	3745.8048	3745.805	Standard Error	0.001961586
6	SALE1565	2025-10-13	CUST1376	PROD14	STORE10	2	675.65	0.2	COD	1081.04	1081.04	Median	0.14
7	SALE74	2025-10-12	CUST1273	PROD20	STORE8	1	184.98	0.1	COD	166.482	166.482	Mode	0.1
8	SALE212	2025-10-12	CUST1442	PROD56	STORE19	4	119.83	0.13	Credit Card	417.0084	417.0084	Standard Deviation	0.087637008
9	SALE418	2025-10-12	CUST1398	PROD62	STORE14	3	969.83	0.02	Credit Card	2851.3002	2851.3	Sample Variance	0.007680245
10	SALE686	2025-10-12	CUST1265	PROD91	STORE9	4	547.21	0.25	Credit Card	1641.63	1641.63	Kurtosis	-1.209177043
11	SALE749	2025-10-11	CUST1325	PROD82	STORE3	3	447.69	0.02	COD	1316.2086	1316.209	Skewness	0.062125565
12	SALE1804	2025-10-11	CUST1241	PROD25	STORE18	1	401.18	0.15	COD	341.003	341.003	Range	0.3
13	SALE345	2025-10-10	CUST1287	PROD50	STORE0	2	660.81	0.18	Credit Card	1083.7284	1083.728	Minimum	0
14	SALE552	2025-10-10	CUST1027	PROD92	STORE11	3	862.18	0.22	PayPal	2017.5012	2017.501	Maximum	0.3
15	SALE1138	2025-10-10	CUST1024	PROD51	STORE6	1	629.09	0.02	Credit Card	616.5082	616.5082	Sum	295.58
16	SALE447	2025-10-09	CUST1253	PROD57	STORE13	3	628.18	0.26	Credit Card	1394.5596	1394.56	Count	1996
17	SALE699	2025-10-09	CUST1059	PROD23	STORE19	3	648.33	0.2	COD	1555.992	1555.992		1
18	SALE710	2025-10-09	CUST1126	PROD31	STORE14	4	471.89	0.12	COD	1661.0528	1661.053		



	A	B	C	D	E	F	G	H	I	J	K
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discount	Payment_Typ	Total_Amou	Colum
23	SALE52	2025-08-25	CUST1148	PROD89	STORE6	3		0.14	Credit Card	486.0633	0

STEP 3:

Calculated the missing Unit price by **Unit_Price = Total Amount / Quantity * (1 - Discount)**



G123 =[@[Total_Amount]] /([@Quantity]* (1-[@Discount]))

	A	B	C	D	E	F	G	H	I	J	K
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discount	Payment_Typ	Total_Amou	Colum
123	SALE52	2025-08-25	CUST1148	PROD89	STORE6	3	188.3966279	0.14	Credit Card	486.0633	486.0633
1064	SALE268	2024-09-20	CUST1406	PROD90	STORE18	3	111.79	0.06	Credit Card	315.2478	315.2478
1572	SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	464.18	0.04	COD	1782.4512	1782.451
2002											
2003											
2004											
2005											
2006											
2007											
2008											
2009											
2010											
2011											

Total Amount =(Quantity * Unit_Price) * (1 - Discount)

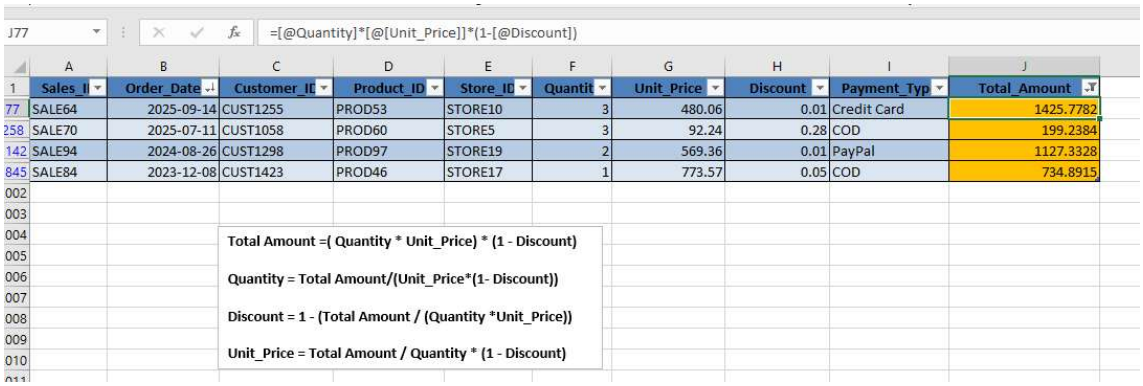
Quantity = Total Amount/(Unit_Price*(1- Discount))

Discount = 1 - (Total Amount / (Quantity *Unit_Price))

Unit_Price = Total Amount / Quantity * (1 - Discount)

STEP 4:

Calculated the missing Total Amount by using Total Amount =(Quantity * Unit_Price) * (1 - Discount)



	A	B	C	D	E	F	G	H	I	J
1	Sales_I	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discount	Payment_Typ	Total_Amount
77	SALE64	2025-09-14	CUST1255	PROD53	STORE10	3	480.06	0.01	Credit Card	1425.7782
258	SALE70	2025-07-11	CUST1058	PROD60	STORE5	3	92.24	0.28	COD	199.2384
142	SALE94	2024-08-26	CUST1298	PROD97	STORE19	2	569.36	0.01	PayPal	1127.3328
845	SALE84	2023-12-08	CUST1423	PROD46	STORE17	1	773.57	0.05	COD	734.8915
002										
003										
004										
005										
006										
007										
008										
009										
010										

STEP 5:

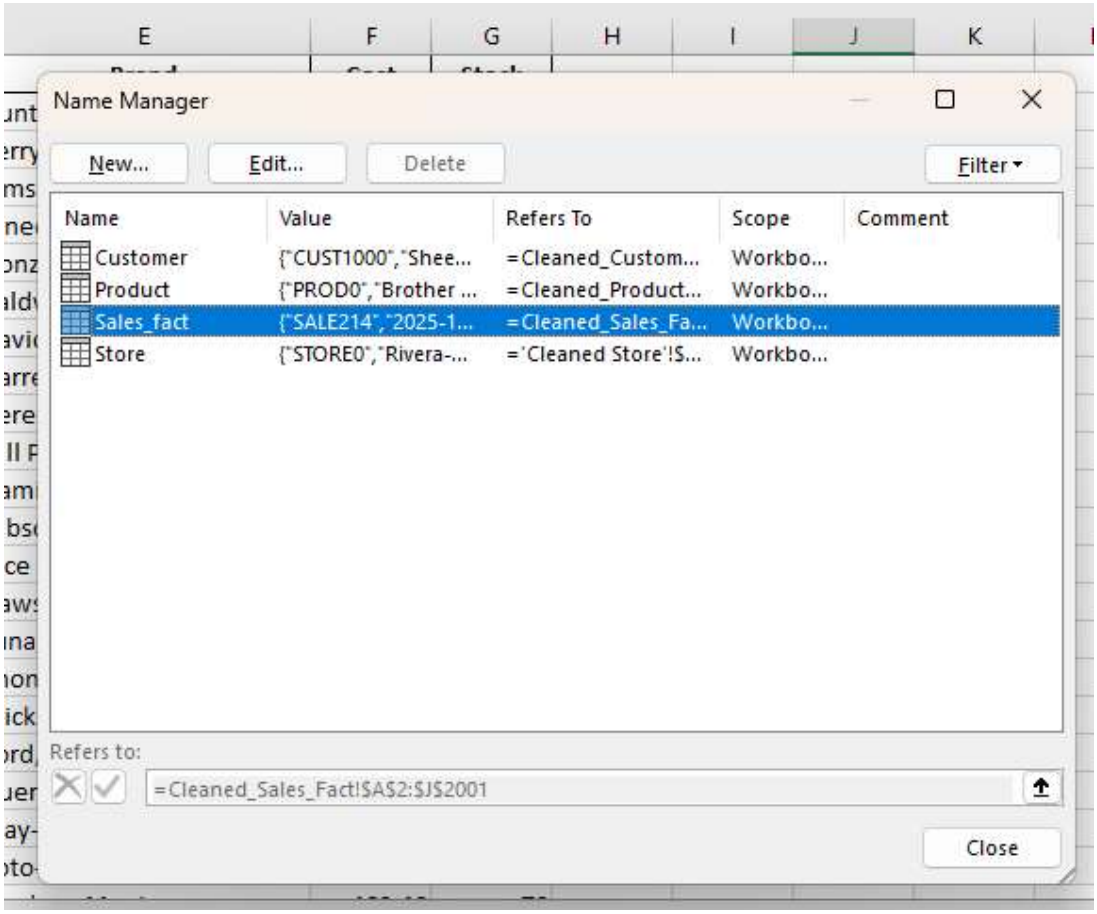
Used Median to fill the missing value in Order Date column



	B	C	D	E	F	G	H	I	J	K	L
1	Order_Date	Customer_ID	Product_ID	Store_ID	Quantit	Unit_Price	Discount	Payment_Typ	Total_Amount		
1755	2024-10-11	CUST1348	PROD99	STORE3	4	451.67	0.15	Credit Card	1535.678	MEDIAN for Order_Date	2024-10-11
2002											

PIVOT TABLE:

Named tables using Naming Manager



Created a **Relationships** between the tables and inserted a pivot tables

Customer ID in both Customer_DIM and Sales _Fact tables

Product ID in bothProduct_DIM and Sales _Fact tables

Store ID in both Store_DIM and Sales _Fact tables

SALES BY CUSTOMER LOYALTY LEVEL

LOYALTY LE	Sum of Total_Amount
Bronze	522623.3819
Gold	517622.0574
Platinum	673616.5011
Silver	460532.7579
Grand Total	**2174394.698**

CATEGORYWISE QUANTITY

CATEGORY	Sum of Quantity
Beauty	738
Clothing	846
Electronics	1165
Home	974
Sports	1241
Grand Total	**4964**

SALES BY CITY

Row Labels	Sum of Total_Amount
Bensonhaven	122176.5471
Christopherland	106248.2833
Danielmouth	101060.6092
East Davidville	109791.7152
East Jenniferstad	99679.086
East Johnstad	110102.399
East Stevechester	104313.0545
Hoodberg	125011.8285
Johnsonville	114401.4715
Lake Michael	93048.1405
New Jimmymouth	111704.9436
New Michaelchester	81343.5889
North Nancyview	98960.8078
Port Josephberg	114533.7966
Port Sharon	118814.1157
Sabrinatown	116829.2206
Thompsonborough	158971.4624
Vasquezborough	94225.7233
Washingtonfurt	85138.5552

SALES BY REGION

Row L	Sum of Total_Amount
East	622404.3163
North	646100.8803
South	635213.0957
West	270676.406
Grand To	**2174394.698**

Pivot Charts:

With the help of pivot table data inserted a pivot charts and added the **slicers**



Used chart titles, style and few other options

Created a column chart,pie chart, bar chart and line chart as above